

Exercise 1.1

$$\mathcal{L} = \mathcal{L}_{\text{QED}} + (D_\mu \phi)^\dagger (D^\mu \phi) - m_\phi^2 \phi^\dagger \phi,$$

where $D_\mu = \partial_\mu + ieA_\mu$ is the covariant derivative and ϕ is a complex scalar field.

We need to get the Feynman rules for the scalar propagator and interaction vertices between photons and scalar particles.

Let's start by expanding $(D_\mu \phi)^\dagger (D^\mu \phi)$ term in the Lagrangian

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{QED}} + [(\partial_\mu + ieA_\mu)\phi]^\dagger [\partial^\mu + ieA^\mu]\phi - m_\phi^2 \phi^\dagger \phi \\ &= \mathcal{L}_{\text{QED}} + (\partial_\mu \phi)^\dagger (\partial^\mu \phi) + (ieA_\mu \phi)^\dagger \partial^\mu \phi + (\partial_\mu \phi)^\dagger (ieA^\mu \phi) + (ieA_\mu \phi)^\dagger (ieA^\mu \phi) - m_\phi^2 \phi^\dagger \phi \end{aligned}$$

$$\Rightarrow \mathcal{L} = \mathcal{L}_{\text{QED}} + \mathcal{L}_\phi \quad \text{with} \quad \mathcal{L}_\phi = \mathcal{L}_{\phi, \text{free}} + \mathcal{L}_{\phi, \text{int}}$$

$$\text{with} \quad \mathcal{L}_{\phi, \text{free}} = (\partial_\mu \phi)^\dagger (\partial^\mu \phi) - m_\phi^2 \phi^\dagger \phi \quad \text{and}$$

$$\mathcal{L}_{\phi, \text{int}} = -ieA_\mu \phi^\dagger (\partial^\mu \phi) + ieA^\mu (\partial_\mu \phi^\dagger) \phi + e^2 A_\mu A^\mu \phi^\dagger \phi$$

We get the ^{scalar} propagator by calculating the two point function:

$$\text{The two point function is } D(x-y) = \langle \Omega | T(\phi(x) \phi^\dagger(y)) | \Omega \rangle$$

and

$$\langle \Omega | T(\phi(x) \phi^\dagger(y)) | \Omega \rangle = \frac{1}{Z_\phi[0,0]} \left(-\frac{i\delta}{\delta J^\dagger_x} \right) \left(-\frac{i\delta}{\delta J_y} \right) Z_\phi[J, J^\dagger] \Big|_{J=J^\dagger=0}$$

Where $Z_\phi[J, J^\dagger]$ is the generating functional given by ^{is conjugate because of term $J\phi^\dagger$ in the exponential}

$$Z_\phi[J, J^\dagger] = \int D\phi D\phi^\dagger \exp \left[i \int d^4x [\mathcal{L}_{\phi, \text{free}} + J\phi^\dagger + J^\dagger \phi] \right]$$

Let's start by shifting the variables to complete the square.

$$\phi \rightarrow \phi - i \int d^4y D_F(x-y) J(y)$$

$\parallel$ $D_F(x-y)$ is the Klein Gordon propagator

$$\phi^\dagger \rightarrow \phi^\dagger + i \int d^4y D_F(x-y) J^\dagger(y)$$

$$D\phi \rightarrow D\phi \quad \text{and} \quad D\phi^\dagger \rightarrow D\phi^\dagger$$

Because a linear shift in ϕ doesn't change the measure $D\phi$.

With this now our partition function becomes $\int_{xy} \equiv \int d^4x d^4y$

$$Z \rightarrow \int D\phi' D\phi'^* \exp\left[i \int d^4x \mathcal{L}_{\text{free}}(\phi')\right] \cdot \exp\left[-i \int_{xy} J_x [-i D(x-y)] J_y\right]$$

Here $\phi' = \phi - i \int d^4y D_F(x-y) J(y) \quad \parallel \phi = \phi(x)$

and $\phi'^* = \phi^* + i \int d^4y D_F(x-y) J^*(y)$

are the shifted variables. $D\phi = D\phi'$ also

independent of ϕ'

Proof by calculation:

$$Z_\phi[J, J^*] = \int D\phi D\phi^* \exp\left[i \int_x \mathcal{L}_0(\phi) + J\phi^* + J^*\phi\right] \quad \parallel \mathcal{L}_0 \equiv \mathcal{L}_{\text{free}}$$

$$\mathcal{L}_0(\phi) = (\partial_\mu \phi^*)(\partial^\mu \phi) - m^2 \phi^* \phi \stackrel{\text{Integration by parts}}{\equiv} -\phi^*(\partial^2 + m^2)\phi$$

And we can write $\phi' = \phi - (\partial^2 + m^2)^{-1} J \quad \parallel \int_y (\partial^2 + m^2) D_F(x-y) J(y) = -i J(x)$
 since D_F is Green's function

For easier computation we write $\partial^2 + m^2 = K \leftarrow \text{operator}$ of KG-operator

$$\Rightarrow Z_\phi[J, J^*] = \int D\phi D\phi^* \exp\left[i \int_x -\phi^* K \phi + J\phi^* + J^*\phi\right]$$

Now we do the shift $\phi \rightarrow \phi + K^{-1} J = \phi' \quad \parallel \begin{cases} -i \int_y D_F(x-y) J(y) = K^{-1} J \\ i \int_y D_F(x-y) J^*(y) = K^{-1} J^* \end{cases}$
 conjugate both sides

$$Z_\phi[J, J^*] = \int D\phi' D\phi'^* \exp\left[i \int_x -(\phi'^* + K^{-1} J^*) K (\phi' + K^{-1} J) + J^*(\phi' + K^{-1} J) + J(\phi'^* + K^{-1} J^*)\right]$$

$$= \int D\phi' D\phi'^* \exp\left[i \int_x -\phi'^* K \phi' - \underbrace{\phi'^* K K^{-1} J}_{=1} - \underbrace{K^{-1} J^* K \phi'}_{=J^*} - \underbrace{K^{-1} J^* K K^{-1} J}_{=J^* K^{-1} J} + \underbrace{J^* \phi'}_{J^*} + \underbrace{J \phi'^*}_{J} + \underbrace{J^* K^{-1} J}_{J^* K^{-1} J} + \underbrace{J K^{-1} J^*}_{J K^{-1} J^*}\right]$$

$$= \int D\phi' D\phi'^* \exp\left[i \int_x \mathcal{L}_0(\phi')\right] \cdot \underbrace{\exp\left[i \int_x J_x K^{-1} J_x^*\right]}_{\text{no } \phi\text{-dependence}} \quad \parallel K^{-1} J_x^* = i \int_y D_F(x-y) J_y^*$$

$$Z_\phi[J, J^*] = \int D\phi' D\phi'^* \exp\left[i \int d^4x \mathcal{L}_0(\phi')\right] \cdot \exp\left[-i \int_{xy} J_x [-i D_F(x-y)] J_y^*\right] \quad \square$$

$$= Z_\phi[0, 0] \cdot \exp\left[-i \int_{xy} J_x [-i D_F(x-y)] J_y^*\right]$$

Now let's calculate the functional derivative to get the propagator

Note on functional derivative: $\frac{\delta J(x)}{\delta J(x)} = \delta(x-y)$ and $\frac{\delta J^*(y)}{\delta J(x)} = 0$
 i.e. J and J^* are independent sources.

$$\begin{aligned}
\langle \Omega | T(\phi(x_1) \phi^*(y_1)) | \Omega \rangle &= \frac{1}{Z_+[0,0]} \left(\frac{-i\delta}{\delta J_{x_1}^*} \right) \left(\frac{-i\delta}{\delta J_{y_1}} \right) Z_+[0,0] \exp \left[+i^2 \int_{x,y} J_x D_F(x-y) J_y^* \right] \Big|_{J=J^*=0} \\
&= i^2 \left(\frac{\delta}{\delta J_{x_1}^*} \right) \left[i^2 \int_y D_F(y, -y) J_y^* \right] \exp \left[i^2 \int_{x,y} J_x D_F(x-y) J_y^* \right] \Big|_{J=J^*=0} \\
&\quad \text{only need to derivate this since } J=J^*=0 \text{ at the end} \\
&= i^4 D_F(y_1, -x_1) = D_F(x_1, y_1) = \langle \Omega | T(\phi(x_1) \phi^*(y_1)) | \Omega \rangle
\end{aligned}$$

Now we use Fourier transformation of $\langle \Omega | T(\phi(x) \phi^*(y)) | \Omega \rangle$ to get $D_F(x-y)$. The Fourier transformation of KG-eq is

$$(-p^2 + m^2) \tilde{D}_F(p) = -i$$

$$\Rightarrow \tilde{D}_F(p) = \frac{-i}{-p^2 + m^2} = \frac{i}{p^2 - m^2}$$

$$\Rightarrow D_F(x-y) = \int \frac{d^4 p}{(2\pi)^4} \tilde{D}_F(p) e^{-ip(x-y)}$$

$$= \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{-ip(x-y)}$$

// $i\epsilon$ prescription when integrating over momenta

$\Rightarrow$ The Feynman rule associated with this propagator is

$$\text{---} \times \text{---} = \frac{i}{p^2 - m^2 + i\epsilon}$$

, which we get directly

From $\langle \Omega | T(\phi(x) \phi^*(y)) | \Omega \rangle = D(x-y)$ by Fourier transforming to momentum space.

We now use operator formalism to find 3-point and 4-point functions

$$\text{The three point function is } \langle \Omega | T(\phi^*(x_1) \phi(x_2) A^\mu(x_3)) | \Omega \rangle = \frac{\langle 0 | T(\phi^*(x_1) \phi(x_2) A^\mu(x_3) e^{i \int x L_{int}}) | 0 \rangle}{\langle 0 | e^{i \int x L_{int}} | 0 \rangle}$$

We expand $e^{i \int x L_{int}}$, to first order in e (only 3-point interaction) we get:

$$e^{i \int x L_{int}} = 1 + i e \underbrace{\int_x [-i A_\mu \phi^* (\partial^\mu \phi) + i A^\mu (\partial_\mu \phi^*) \phi]}_{\text{Interaction term for us}} + \dots$$

Now we use Wick's theorem to find possible contractions:

$$\langle 0 | T(\phi^*(x_1) \phi(x_2) A^\mu(x_3) e^{\int_x A_\mu (\phi^* (\partial^\mu \phi) - (\partial^\mu \phi^*) \phi)}) | 0 \rangle = \langle 0 | \text{all possible contractions} | 0 \rangle$$

Through LSZ theorem we know that only 1PI-diagrams contribute so let's find the contraction for that along with any possible symmetry factor

The only 1-PI contraction is

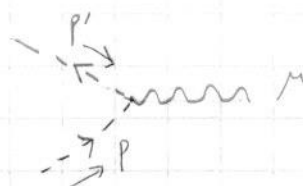
$$e \phi^*(x_1) \phi(x_2) A^\mu(x_3) \int_x A_\mu (\phi^* (\partial^\mu \phi) - (\partial^\mu \phi^*) \phi)$$

These are the only contractions since we cannot have contractions like

$$\underbrace{\phi^* \phi} \quad \text{and} \quad \underbrace{\phi^* A_\mu}$$

Since we don't have terms in $\uparrow$ Lagrangian for interaction $--\text{wavy}--$ and $--\rightarrow \leftarrow --$ two point functions.

$\Rightarrow$ The only allowed vertex is



in position-space the Feynman rule is

$$-ie \int d^4x [(\partial^\mu \phi) - (\partial^\mu \phi^*)] \quad \text{which in momentum space gives}$$

$$\underline{-ie (p - p')^\mu} = \text{diagram} \quad \parallel \text{ if we reverse } p' \text{ direction in diagram we get } -ie (p + p')^\mu$$

Now we look for 4-point function. The expansion of $e^{i \int \mathcal{L}_{int}}$ including the 4-point interaction to the lowest order is:

$$e^{i \int \mathcal{L}_{int}} = 1 + ie \int_x -ie A_\mu \phi^* \partial^\mu \phi + ie A_\mu (\partial^\mu \phi^*) \phi + ie^2 \int_x A_\mu A^\mu \phi^* \phi$$

The 4-point interaction part is $ie^2 \int_x g_{\mu\nu} A_\mu A_\nu \phi^* \phi$

$\Rightarrow \langle 0 | T(\phi^*(x_1) \phi(x_2) A^\mu(x_3) A^\nu(x_4) \int_x ie^2 g_{\mu\nu} A_\mu A_\nu \phi^* \phi) | 0 \rangle$ is the numerator part of 4-point function for only 4-point int

There are two possible ways to contract A's which gives us factor of 2

$\Rightarrow$ $= \underline{2ie^2 g_{\mu\nu}}$ is now the Feynman rule

Exercise 1.2.

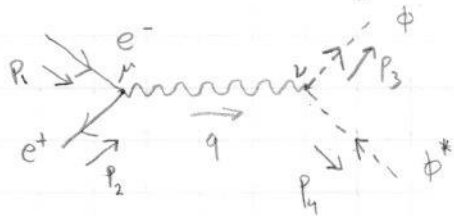
Compute the differential cross-section $d\sigma/d\Omega$ for scattering process $e^+e^- \rightarrow \phi^*\phi$ assuming $m_\phi \gg m_e$ and that the incoming particles are unpolarized.

Let's start by drawing possible Feynman diagrams

From Lagrangian $\mathcal{L} = \mathcal{L}_{\text{QED}} + (D_\mu \phi)^\dagger (D^\mu \phi) - m\phi^*\phi$ we see

that electron and the ϕ -fields are not coupled so interactions must happen via mediating photons. Since the final product is 2-particles ϕ and ϕ^* we only need to consider 3-point interaction vertices

The only available Feynman diagram is:



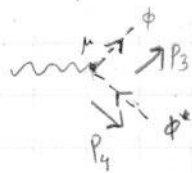
From QED we know the photon propagator to be

$$\frac{-i g_{\mu\nu}}{q^2}, \quad \mathcal{L}_{\text{QED}} \text{ also gives vertex } \not{A} = -ie \gamma^\mu$$

The external legs for scalar particles give

1 and for Dirac-particles it gives Dirac spinor or adjoint spinor (u^s and $\bar{u}^s$)

The vertex



has Feynman rule $-ie(-p_3 + p_4)^\mu = ie(p_3 - p_4)^\mu$

Using these Feynman rules we get the matrix element

$$-i\mathcal{M} = \bar{u}^r(p_2) [-ie\gamma^\mu] u^s(p_1) \frac{-ig_{\mu\nu}}{q^2} 1 \cdot [ie(p_3 - p_4)^\nu] \cdot 1$$

$$\mathcal{M} = \frac{e^2}{q^2} \bar{u}^r(p_2) \gamma^\mu u^s(p_1) (p_3 - p_4)_\mu$$

The matrix element squared is:

$$\Rightarrow \mathcal{M}\mathcal{M}^* = \frac{e^2}{q^2} \bar{u}^r(p_2) \gamma^\mu u^s(p_1) (p_3 - p_4)_\mu \bar{u}^s(p_1) \gamma^\nu u^r(p_2) (p_3 - p_4)_\nu = |\mathcal{M}|^2$$

Now we average the initial spins since initial particles are unpolarized:

$$\langle |\mathcal{M}|^2 \rangle = \frac{1}{4} \sum_{sr} |\mathcal{M}|^2$$

Now let's write $|\mathcal{M}|^2$ in index form to make use of identity $\sum_s u^\alpha(p) \bar{u}^\beta(p) = \not{p} + m$ since $m_\phi \gg m_e \Rightarrow m_e \approx 0$ can be approximated.

$$\Rightarrow \langle |\mathcal{M}|^2 \rangle = \frac{e^2}{4q^2} \sum_{sr} \underbrace{\bar{u}_a^r(p_2) \gamma^\mu_{ab} u_b^s(p_1)}_{\text{Trace}} \bar{u}_c^s(p_1) \gamma^\nu_{cd} u_d^r(p_2) (p_3 - p_4)_\mu (p_3 - p_4)_\nu$$

$$= \frac{e^2}{4q^2} \left[\sum_s u_b^s(p_1) \bar{u}_c^s(p_1) \right] \gamma^\nu_{cd} \left[\sum_r u_d^r(p_2) \bar{u}_a^r(p_2) \right] \gamma^\mu_{ab} (p_3 - p_4)_\mu (p_3 - p_4)_\nu \quad \parallel \text{We can recognize trace from indices.}$$

$$\Rightarrow \langle |M|^2 \rangle = \frac{e^2}{4q^2} \text{Tr}[\not{p}_1 \gamma^\nu \not{p}_2 \gamma^\mu] (p_3 - p_4)_\mu (p_3 - p_4)_\nu$$

$$= \frac{e^2}{4q^2} \text{Tr}[\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] p_{1\mu} p_{2\rho} (p_3 - p_4)_\sigma (p_3 - p_4)_\nu$$

Let's rename the indices abit
for convenience later

Now we use the trace identity (Peskin Chapter 5.1)

$$\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] = 4(g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho})$$

$$\Rightarrow \langle |M|^2 \rangle = \frac{e^2}{q^2} [g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}] p_{1\mu} p_{2\rho} (p_3 - p_4)_\sigma (p_3 - p_4)_\nu$$

$$= \frac{e^2}{q^2} [(p_1 \cdot (p_3 - p_4))(p_2 \cdot (p_3 - p_4)) - (p_1 \cdot p_2)(p_3 - p_4)^2 + (p_1 \cdot (p_3 - p_4))(p_2 \cdot (p_3 - p_4))]$$

$$\langle |M|^2 \rangle = \frac{e^2}{q^2} [2[p_1 \cdot (p_3 - p_4)][p_2 \cdot (p_3 - p_4)] - (p_1 \cdot p_2)(p_3 - p_4)^2]$$

$$q^2 = s = (p_1 + p_2)^2 = 2(p_1 \cdot p_2) \quad , \quad \text{where } s \text{ is the Mandelstam } s$$

Let's now setup the kinematics. Let's calculate the cross section in centre of mass (CM) frame:

$$\text{In this frame we have } \vec{p}_1 = -\vec{p}_2, E_1 = E_2, \vec{p}_3 = -\vec{p}_4, E_3 = E_4$$

$$\text{Also } s = (p_1 + p_2)^2 = (p_3 + p_4)^2 = 4E_3^2 = 4E_1^2 \quad \text{and } |\vec{p}_1|^2 = E_1^2 \quad \text{and } E_3^2 = |\vec{p}_3|^2 + m_\phi^2$$

$$\Rightarrow (p_3 - p_4) = (0, 2\vec{p}_3) \Rightarrow p_1 \cdot (p_3 - p_4) = -\vec{p}_1 \cdot 2\vec{p}_3 \quad \text{and } p_2 \cdot (p_3 - p_4) = \vec{p}_1 \cdot 2\vec{p}_3$$

$$\Rightarrow \underline{2(p_1 \cdot (p_3 - p_4))(p_2 \cdot (p_3 - p_4))} = 2(-4)(\vec{p}_1 \cdot \vec{p}_3)^2 \quad \text{if we define } \theta \text{ to be angle between } \vec{p}_1 \text{ and } \vec{p}_3 \text{ we get } \vec{p}_1 \cdot \vec{p}_3 = |\vec{p}_1||\vec{p}_3|\cos\theta$$

$$\Rightarrow -8(\vec{p}_1 \cdot \vec{p}_3)^2 = -8|\vec{p}_1|^2 |\vec{p}_3|^2 \cos^2\theta \quad || \vec{p}_1|^2 = E_1^2 = \frac{s}{4}, \quad |\vec{p}_3|^2 = \frac{s}{4} - m_\phi^2$$

$$\Rightarrow \underline{2(p_1 \cdot (p_3 - p_4))(p_2 \cdot (p_3 - p_4))} = -8 \frac{s}{4} (\frac{s}{4} - m_\phi^2) \cos^2\theta = 2s(m_\phi^2 - \frac{s}{4}) \cos^2\theta$$

$$\text{Now } (p_3 - p_4)^2 = (0, 2\vec{p}_3)^2 = -4|\vec{p}_3|^2, \quad -(p_1 \cdot p_2) = -\frac{1}{2}s$$

$$\Rightarrow -(p_1 \cdot p_2)(p_3 - p_4)^2 = 2s|\vec{p}_3|^2 = 2s(\frac{s}{4} - m_\phi^2)$$

$$\Rightarrow \langle |M|^2 \rangle = \frac{e^2}{s} [2s(m_\phi^2 - \frac{s}{4}) \cos^2\theta + 2s(\frac{s}{4} - m_\phi^2)]$$

$$= \frac{e^2}{s} [2s(\frac{s}{4} - m_\phi^2)(1 - \cos^2\theta)]$$

$$\Rightarrow \langle |M|^2 \rangle = 2e^2 \left[\frac{s}{4} - m_\phi^2 \right] (1 - \cos^2 \theta)$$

Now we need to get the cross section using this Amplitude squared:

The cross section is given by $\frac{d\sigma}{d\Omega} = \frac{\langle |M|^2 \rangle}{64\pi^2 s} \frac{p_f^*}{p_i^*}$ for $2 \rightarrow 2$ scattering in CM-frame

p_f^* is final state momentum and

p_i^* is initial state momentum $\Rightarrow p_f^* = |\vec{p}_3|$ and $p_i^* = |\vec{p}_1| = E_1 = \sqrt{\frac{s}{4}}$

$$|\vec{p}_3| = (E_3^2 - m_\phi^2)^{1/2} = \left(\frac{s}{4} - m_\phi^2 \right)^{1/2}$$

$$\Rightarrow \frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{\sqrt{s/4 - m_\phi^2}}{\sqrt{s/4}} 2e^2 \left[\frac{s}{4} - m_\phi^2 \right] (1 - \cos^2 \theta)$$

$$\frac{d\sigma}{d\Omega} = \frac{e^2}{8\pi^2} \frac{(s/4 - m_\phi^2)^{3/2}}{(s/4)^{3/2}} (1 - \cos^2 \theta) \quad \parallel \quad \frac{e^2}{4\pi} = \alpha$$

$$\frac{d\sigma}{d\Omega} = \frac{\alpha}{2\pi} \left[\frac{s - 4m_\phi^2}{s} \right]^{3/2} (1 - \cos^2 \theta)$$

Exercise 1.3

Vacuum polarisation at 1-loop:

For spinor QED-process we have one diagram:



Now we add the scalar QED-process too:

At order of 1-loop we get



and



3-vertices at the ends of loop

4-vertices at the beginning/end of loop

$\Rightarrow$ We get extra term compared to spinor-QED since we have a 4-point interaction

$$\Rightarrow i\Pi^{\mu\nu}(k) = \text{diagram with 3-vertices} = \text{diagram with 4-vertices} + \text{diagram with 4-vertices} + \text{diagram with 4-vertices}$$

$$i\Pi^{\mu\nu}(k) = (-1) \int \frac{d^4 p}{(2\pi)^4} \text{Tr} [(-ie\gamma^\mu) \frac{i}{\not{p} + \not{k} - m} (ie\gamma^\nu) \frac{i}{\not{p} - m}] \quad \leftarrow \text{electron loop}$$

$$+ \int \frac{d^4 q}{(2\pi)^4} (ie[q+k-q]^\mu) \frac{1}{(q+k)^2 - m_\phi^2 + i\epsilon} (-ie[q+k-q]^\nu) \frac{1}{q^2 - m_\phi^2 + i\epsilon}$$

$$+ \int \frac{d^4 l}{(2\pi)^4} (2ie^2 g^{\mu\nu}) \frac{1}{l^2 - m_\phi^2 + i\epsilon}$$

So we get

$$i\pi^{\mu\nu}(k) = e^2 \int \frac{d^4 p}{(2\pi)^4} \text{Tr} \left[\gamma^\mu \frac{i}{\not{p} + \not{k} - m} \gamma^\nu \frac{i}{\not{p} - m} \right] \\ + e^2 k^\mu k^\nu \int \frac{d^4 q}{(2\pi)^4} \frac{1}{(q+k)^2 - m_\phi^2 + i\varepsilon} \frac{1}{q^2 - m_\phi^2 + i\varepsilon} \\ + 2ie^2 g^{\mu\nu} \int \frac{d^4 \ell}{(2\pi)^4} \frac{1}{\ell^2 - m_\phi^2 + i\varepsilon}$$

The expression for only $\mathcal{L}_\phi$ -theory would then be

$$i\pi_\phi^{\mu\nu}(k) = e^2 \left(2i g^{\mu\nu} \int \frac{d^4 \ell}{(2\pi)^4} \frac{1}{\ell^2 - m_\phi^2 + i\varepsilon} + k^\mu k^\nu \int \frac{d^4 q}{(2\pi)^4} \frac{1}{(q+k)^2 - m_\phi^2 + i\varepsilon} \frac{1}{q^2 - m_\phi^2 + i\varepsilon} \right)$$

or something of similar form.

Exercise 1.4

From $i\pi^{\mu\nu}(k)$ in previous exercise we see that the only tensors that can appear in

$\pi^{\mu\nu}(q)$ are $g^{\mu\nu}$ and $q^\mu q^\nu$. Ward identity tells us

$$q_\mu \pi^{\mu\nu}(q) = 0 \Rightarrow \pi^{\mu\nu}(q) \propto \left(g^{\mu\nu} - \frac{q^\mu q^\nu}{q^2} \right)$$

Since $q_\mu \left(g^{\mu\nu} - \frac{q^\mu q^\nu}{q^2} \right) = q^\nu - q^\nu = 0$

So $\pi^{\mu\nu}(q)$ must be proportional to $\left(g^{\mu\nu} - \frac{q^\mu q^\nu}{q^2} \right)$

Since $\pi^{\mu\nu}(q)$ doesn't have pole at $q^2=0$ (we can see this by the expression we got for $i\pi^{\mu\nu}(k)$ in previous exercise)

There should be no pole in the new form we trying to morph it into

Multiply the proportionality by q^2
 $\Rightarrow \pi^{\mu\nu}(q) \propto (g^{\mu\nu} q^2 - q^\mu q^\nu)$ and we set proportionality factor to be $\pi(q^2)$ (no poles in this either)

$\Rightarrow$ So with these arguments we can write

$$\pi^{\mu\nu}(q^2) = (g^{\mu\nu} q^2 - q^\mu q^\nu) \pi(q^2) \quad \square$$